Lab 2: Decomposition

Reference solution

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2025–2026

Reference Python code is in `correction.py`; this document gives the analytical answers and expected numerical values.

Exercise 1 – Synthetic decomposition

Solution

The 4-panel figure shows the observed series with a clear linear drift and a regular 12-month sinusoidal pattern of constant amplitude 5 — the visual signature of an *additive* model.

Exercise 2 – Trend estimation: three methods

Solution

Reference RMSE values (over $T = 240$, seed 42):

Method	**RMSE vs. true trend**
Centred MA, $k = 6$	≈ 0.39
Linear regression (correctly specified)	≈ 0.07
LOESS, frac $= 0.2$	≈ 0.31

The linear regression is the best estimator here because it matches the true generating model. The MA loses a few observations at each end and absorbs some sinusoidal residual at length 13. LOESS provides flexibility but adds variance. The polynomial estimate is a clean straight line; LOESS wiggles slightly around it.

Exercise 3 – Seasonality estimation: three methods

Solution

After LOESS detrending, the three methods produce visually similar estimates of the 12-month seasonal pattern. RMSE versus the truth $5\sin(2\pi k/12)$ (seed 42):

- Seasonal averaging: ≈ 0.31.
- Dummy regression (11 dummies): ≈ 0.31 (mathematically identical to averaging when no other regressor is present).
- Cosine–sine model (1 harmonic): ≈ 0.16 — **best**, using only 2 parameters.

The Fourier model wins here because the truth is a *pure* sinusoid, exactly captured by the 1-harmonic basis (which lets the estimator pool all 240 observations into 2 coefficients). Averaging and dummies estimate 12 independent indices, with one degree of freedom per index and therefore higher variance. With more harmonics ($J \geq 2$), the Fourier model can fit more complex seasonal shapes; with all $J = \lfloor d/2 \rfloor$ harmonics it is equivalent to the dummies (Proposition 2.5 of the lecture).

Exercise 4 – Application to Atlanta temperatures

Solution

(1) Period $d = 12$ (annual seasonality). The amplitude of the seasonal swing looks roughly constant over the 138-year span, supporting an additive model.
(2) Visual decomposition: the trend (centred MA, $k = 6$) sits around 61.8°F. The 12 monthly seasonal indices, in °F, are

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
−19.9	−17.1	−9.5	−0.3	+8.7	+16.2	+18.6	+17.7	+12.2	+1.2	−9.9	−17.9

Peak in July (+18.6), trough in January (−19.9), peak-to-peak amplitude $\sim$ 38°F. The residual has std $\approx$ 2.9°F — mostly weather noise.
(3) `statsmodels.seasonal_decompose(y, period=12, model="additive")` gives the same monthly seasonal indices (within floating-point tolerance) when using the same 2×12 MA filter.

Exercise 5 – Bonus: additive vs multiplicative

Solution

The multiplicative simulated series shows a seasonal amplitude that *grows* with the trend level. The additive workflow on Y_t leaves a clearly heteroscedastic residual (funnel-shaped), whereas the same workflow on $\log Y_t$ gives an approximately stationary residual.
Conclusion: prefer the multiplicative model (or work on $\log Y_t$) when the seasonal amplitude scales with the level — typical of sales, GDP, population, financial volume.